A randomised controlled trial of a pharmaceutical care programme in high-risk diabetic patients in an outpatient clinic

RHONDA M. CLIFFORD, KEVIN T. BATTY, TIMOTHY M. E. DAVIS, WENDY DAVIS, GARY STEIN, GRAEME STEWART and RICHARD J. PLUMRIDGE

Objective — To determine the impact of a pharmaceutical care programme (PCP) in diabetic patients.
Design — Randomised controlled study of high-risk diabetic patients.
Setting — Outpatient clinic at Fremantle hospital (FH), Western Australia.
Method — Patients over 18 years of age who could communicate freely in English and fulfilled pre-determined criteria for being high-risk for the development of diabetic complications, were randomly assigned to the PCP or control groups in a ratio of 2:1. In the PCP arm, a clinical pharmacist reviewed and monitored all aspects of the patients' drug therapy in collaboration with other health care professionals at six-weekly intervals for six months. The control patients received usual outpatient care.
Main outcome measures — Glycosylated haemoglobin (HbA_{1c}), quality of life (QOL), patient satisfaction with health care providers and changes in drug therapy during the PCP.
Results — Seventy-three patients were recruited into the study, of whom 48 (66 per cent) were randomised to the PCP. There were no significant differences between the PCP (cases) and the control groups for demographic variables. The mean (±SD) HbA_{1c} for the cases was 8.4±1.4 per cent at the beginning and 8.2±1.5 per cent at the end of the study period ($P>0.05$). There was similarly no change in the control group (8.5±1.6 per cent to 8.1±1.6 per cent; $P>0.05$). There were no significant changes in QOL for cases or controls over the period of the study. During the PCP, there was a significant increase in patient satisfaction with the care provided by the clinical pharmacist ($P=0.007$) and the provision of drug information ($P=0.036$). The clinical pharmacist facilitated 39 drug interventions in the 48 cases. A high level of complementary medicine usage was found in the PCP group (16.7 per cent).
Conclusions — PCPs provide patients with important medication information and result in changes to drug therapy. However, in diabetic patients under specialist care, a six-month PCP did not lead to an improvement in glycaemic control. The role for pharmacist intervention in primary care now needs to be evaluated.

DIABETES mellitus is increasing in prevalence and is commonly associated with prolonged ill health and premature death.[1,2,3] The morbidity rate in patients with diabetes may be up to 11 times higher than in persons without the disease.[4] Recent large-scale studies in both type 1 and type 2 diabetes have shown that the risk of vascular complications can be reduced through intensive treatment aimed at normalising glycaemia.[4-6]

Pharmaceutical care (PC) is the process through which a clinical pharmacist co-operates with a patient and other health professionals in designing, implementing, and monitoring a therapeutic plan that will produce specific outcomes.[7,8] This includes identifying potential and actual drug-related problems, resolving current drug-related problems, and preventing future drug-related problems. Reports of pharmaceutical care programmes (PCPs) in an outpatient setting have generally shown improvement in

School of Pharmacy, Curtin University of Technology, Australia
Rhonda M. Clifford, Grad Dip Pharm, *PhD candidate*
Kevin T. Batty BPharm PhD, *senior lecturer*

Department of Medicine, University of Western Australia
Timothy M. E. Davis, DPhil, FRACP, *professor*

Fremantle Hospital and Health Service
Wendy Davis, MPH, *research officer, department of medicine*
Gary Stein MBBS, FRACP, *consultant physician*
Graeme Stewart, MBBS, FRACP, *consultant physician*
Richard J. Plumridge, MPA, FPS, *director of pharmacy*

Correspondence: Ms Clifford, Senior Pharmacist, Fremantle Hospital and Health Service, Alma St, Fremantle 6160, Australia
rhonda.clifford@health.wa.gov.au

Int J Pharm Pract 2002:10:85-9

compliance, patient knowledge and relatively short-term indicators of improved pharmacotherapy, such as the international normalised ratio (INR) in studies of warfarin therapy.[9-11] In a recent study of a PCP in diabetes, eligible outpatients were randomised to either a PCP or control group and followed over a four-month period. Patients in the intervention group received diabetes education, drug counselling, instructions on dietary regulation, exercise, home blood glucose monitoring, and evaluation and adjustment of their hypoglycaemic regimen. Although a statistically significant decrease in HbA_{1c} was seen in the intervention group, the patients remained poorly controlled, with glycosylated haemoglobin (HbA_{1c}) greater than 9 per cent.[12]

The aim of the present study was to determine the impact of a PCP in diabetic patients attending a hospital outpatient clinic. We hypothesised that a PCP would improve glycaemic control without impacting adversely on the quality of life of the patient or their satisfaction with the health care provided. The specific outcome measures that were used to determine the impact of the programme were:

- Glycaemic control as assessed by HbA_{1c}
- Quality of life (QOL) as measured by a diabetes-specific quality of life questionnaire
- Patient satisfaction with health care providers and drug information provided, as measured by a questionnaire specifically developed for the study.

Methods

We recruited adult patients aged over 18 years with either type 1 or type 2 diabetes and at least one of the following features indicating the patient was high-risk for the development of diabetes complications:

1. Random blood glucose levels >11mmol/L on at least two occasions in a tertiary care setting in the previous 12 months
2. HbA_{1c} > 8 per cent on at least two occasions in the previous 12 months
3. Hypertension (systolic blood pressure [SBP] >160mmHg and/or diastolic blood pressure [DBP] >90mmHg) and/or taking drug therapy
4. Dyslipidaemia (total serum cholesterol >5.5mmol/L and/or serum triglycerides >4.0mmol/L)
5. Polypharmacy (> three drugs)

The study was carried out in the Fremantle hospital (FH) diabetes outpatient clinic and was approved by the hospital's ethics committee and the Curtin University human research ethics committee. Patients were followed for six months from recruitment. During the recruitment phase (October 1998-April 1999), the medical records of all scheduled outpatients were reviewed for eligibility one week prior to the clinic appointment. Patients who met the inclusion criteria were then telephoned to determine their willingness to participate in the study and asked to attend their scheduled outpatient appointment 45 minutes early. All patients who were telephoned agreed to participate in the study. Patients were then allocated to case or control groups using a 2:1 computer-generated randomisation schedule.

On the day of enrolment, patients were provided with a written information sheet explaining the study, and were given a verbal explanation from the investigator, and they subsequently signed an informed consent form. Demographic information and data relating to glycaemic control over the previous six months were obtained from each patient's case notes. Specific outcomes included change in HbA_{1c}, change in quality of life and patient satisfaction over time using the instruments described below.

All patients completed a modified version of the Diabetes Quality of Life (DQOL) instrument developed for the Diabetes Control and Complications Trial (DCCT).[13,14] The DQOL survey was divided into three sections: (i) 10 questions related to the patient's general satisfaction (satisfaction); (ii) 10 questions related to how often the patient's diabetes interfered with their life (impact); and (iii) six questions related to how often the patient worried about their diabetes (worry). Responses to DQOL questions were rated on a Likert scale from 1 to 5, with higher scores indicating greater dissatisfaction, worry or impact of their diabetes. For each subscale and the total, individual item scores were averaged.

The patient satisfaction survey consisted of 20 questions related to patient satisfaction with the hospital physician and clinical pharmacist, including satisfaction with the information provided by the clinical pharmacist relating to drug therapy. The survey was developed for the study and was pre-tested on a random sample of 25 patients from the pharmacy outpatient section to ensure that it was easy to administer and complete. Patients were asked to rate their satisfaction with the hospital physician, clinical pharmacist and the hospital drug information provided on a scale from 1 to 4 (very dissatisfied through to very satisfied). An independent person administered the survey in each case.

All patients randomised to the PCP attended appointments with an experienced clinical pharmacist at six-weekly intervals for six months. The same clinical pharmacist, who had completed a comprehensive, self-directed revision of diabetes management prior to the study, saw each patient at every visit, eliminating bias that may have arisen from using different clinical pharmacists. A comprehensive list of questions relating to pharmacotherapy and diabetes, including use of proprietary and non-proprietary medicines (including complementary medicines), was covered during the patient's appointment with the clinical pharmacist. The PCP was car-

Table 1: Patient demographics	Cases	Controls
Total number	48	25
Type 1 diabetes (number)	14	5
Type 2 diabetes (number)	34	20
Age (years)	60±12	61±12
Male	28 (58%)	12 (48%)
HbA_{1c} (%)	8.4±1.4	8.5±1.6

Table 3: Categories of interventions (n=39)	
Intervention	Number (%)
Addition of a drug	16 (41)
Cessation of a drug	8 (20)
Adverse drug reaction	1 (3)
Drug/food interaction	9 (23)
Compliance	2 (5)
Dose increase	3 (8)

ried out in co-operation with the diabetes physicians and other members of the diabetes health care team, who were aware of the PCP/control allocation for patients. The control patients received standard outpatient care for their diabetes but no intervention from the clinical pharmacist. These patients completed the QOL survey but not the patient satisfaction survey. All patients had their HbA_{1c} measured at the beginning and end of the study.

Based on a previous study,[12] we determined that 17 PCP patients and 17 controls were required to demonstrate a 20 per cent difference in the HbA_{1c} at the end of the study (power > 80 per cent, α=0.05). To account for potential withdrawals (particularly from the PCP group) and to increase the power of the study, the target was 50 PCP patients and 25 controls.

Statistical analyses were performed using the Statistical Package for Social Sciences (SPSS) program. Statistical differences in continuous outcome measures between the two groups were assessed using Student's t-test for independent samples. Baseline and endpoint outcome measures were compared using the paired t-test. Categorical data were assessed by chi-square tests. QOL and patient satisfaction scores were similarly analysed. All results are expressed as mean ± standard deviation (SD). A significance level of <0.05 was used.

Results

Seventy-three patients were recruited: 48 cases and 25 controls (Table 1). All patients were included in the subsequent primary outcome measures analysis. There were no significant differences between cases and controls for any of the demographic variables that were documented for all patients. As hypertension, dyslipidaemia and polypharmacy were inclusion criteria that did not apply to all patients, these data were not analysed further.

During the six-month study period the mean (±SD) HbA_{1c} levels did not change significantly in either the cases (8.4±1.4 per cent at the beginning and 8.2±1.5 per cent at the end) or the controls (8.5±1.6 per cent and 8.1±1.6 per cent) (P>0.2 across time and between groups). Similarly, there were no significant differences in QOL within or between groups during the course of the study (P>0.15): 2.0±0.6 and 1.9±0.5 for the cases and 1.9±0.5 and 1.9±0.4 for the controls at the beginning and the end respectively.

Table 2: Results of the patient satisfaction survey (a higher score indicates a higher degree of satisfaction; data are presented as mean ± SD)	Day one	Six months
Hospital physician satisfaction	3.1±0.6	3.3±0.8
Clinical pharmacist satisfaction	2.9±0.6	3.3±0.8*
Drug information satisfaction	2.9±0.5	3.1±0.5*

*P<0.05 for baseline vs six months

Table 4: Use of complementary medicines (CMs)	
Cases (n=48)	Controls (n=25)
Number taking CMs=8 (16.7%)	Number taking CMs=4 (16%)
Number of CMs=8	Number of CMs=6
1. **Omega-3 fish oil**	1. Herbalax
2. **Chromium**	2. Linseed and **garlic**
3. **Ginkgo biloba**	3. Grape seed extract
4. **Gotu-cola**	4. **Fish oil** and **glucosamine**
5. **Glucosamine**	
6. **Cod liver oil**	
7. Thiamine	
8. Thiamine	

CMs in **bold** had the potential to interfere with another medication or affect an existing disease state in the individual patient

The results of the patient satisfaction survey are summarised in Table 2. While the baseline score for the pharmacist was lower than that for the physician, there was a significantly higher satisfaction with the clinical pharmacist at the end of the study than at the beginning (P=0.007), as well as a greater patient satisfaction with drug information (P=0.036).

Thirty-nine drug therapy interventions were made in the PCP in 39 different patients after a full medication review and subsequent discussion with the consultant physician (Table 3). All drug therapy suggestions that were discussed with the consultant physician were implemented. No interventions were made in the control group as no medication review was carried out in this group by the clinical pharmacist (although baseline drug use was recorded). During the study it was determined that 16 per cent of patients were taking at least one complementary medicine, usually without the knowledge of the consultant physician. Fourteen different complementary medicines were being consumed and nine of these (64 per cent) had the potential to interfere with another medication or affect an existing disease state in the individual patient (Table 4).

Discussion

We found that the PCP was successfully accommodated in an outpatient clinic setting and was

well received by patients, doctors and other health care professionals. The PCP did not adversely affect the patient's QOL and it significantly improved patient satisfaction both with the care provided by the clinical pharmacist and with provision of drug-related information. The PCP was not associated with any improvement in glycaemic control within the six-month study period, as assessed by HbA_{1c}. A longer time frame and increased intensity of intervention may be required to show a clinical improvement in glycaemic control in a tertiary care setting.

We targeted high-risk patients on the basis that they would have the greatest potential for clinical benefit from the PCP. It is also possible that the multiple problems and complexities of care facing these patients blunted the impact of the PCP. In addition, these patients are usually reviewed by up to five different health professionals after referral by their general practitioner to the clinic, including a consultant physician. Even with this level of care, HbA_{1c} levels below 8 per cent were not achieved in most of the case and control subjects. The value of adding a pharmacist in this context appears limited. Nevertheless, the medication review process is an important part of the management of the diabetic patient. The clinical pharmacist resource may be better utilised in the primary care setting. In Australia, patients receive medication counselling at the point of dispensing; however, time constraints may prevent a full medication review in this situation. Intervention in this setting needs to be investigated in future studies.

There has been one previous small-scale study of PCP in a diabetes outpatient clinic involving 39 urban African-American patients.[12] A range of interventions was implemented in addition to medication counselling. The mean HbA_{1c} at the beginning of the study was >12 per cent and it was reduced to 9.2 per cent (there was no control group). Given the very high initial HbA_{1c} levels in this cohort, it is not surprising that a positive result was achieved. However, recent evidence suggests that there is no threshold HbA_{1c} level for vascular benefit in diabetes.[15] Although our data from a restricted sample do not show a significant reduction in HbA_{1c} even a small reduction as a result of a PCP may prove beneficial.

Thirty-nine interventions were implemented in the case group. These interventions could involve medications other than those for diabetes. While the interventions did not result in a significant improvement in glycaemic control, it may be that additional vascular risk factors, such as blood lipids and blood pressure, were improved. These secondary endpoints were not considered in the study.

During the process of medication review, one in six patients were found to be consuming one or two complementary medicines. Complementary medicines or "herbal" preparations are being used by an increasing number of patients who do not usually advise their health care providers of concomitant use.[16-18] In Australia, complementary medicines range from government registered products in pharmacies and health food shops to completely uncontrolled "herbal" preparations produced in-house by Chinese herbalists and naturopaths.[19,20] Drug-herb and disease-herb interactions can occur and some complementary medicines have been linked to severe adverse reactions.[18,21-23] The majority of our patients who were taking complementary medicines were considered to be at risk of such interactions and adverse effects.

The present study highlights the value of the pharmacist as an information resource for patients with diabetes. This extends to complementary medicines where potential interactions with conventional therapy may be neither suspected nor recognised. Our data and those of others suggest that a larger evaluation of a PCP for diabetes in the primary care setting may be valuable, a situation in which the cost-effectiveness of a diabetes-specific PCP could be evaluated.

Conclusion

Our study demonstrated that, in a hospital outpatient clinic that treats diabetic patients with higher than average risk factors, the use of a PCP had no impact on glycaemic control when undertaken over a short period. The medication review process did result in drug therapy changes and the programme highlighted the need to elucidate a full medication history, including the use of complementary medicines. We conclude that the medication review process is an important part of the care of the diabetic patient; however, the pharmacist resource may be better utilised in a primary care setting and this warrants further investigation.

ACKNOWLEDGMENTS: This study was funded by The Society of Hospital Pharmacists of Australia, Pharmacia and Upjohn Contemporary Therapeutics Research Grant.

References

1. Patel A. Diabetes in focus. London: Pharmaceutical Press; 1999.
2. Bloomgarden ZT. The European Association for the Study of Diabetes annual meeting, 1998. Diabetes Care 1999;22:989-95.
3. Cost of diabetes. Alexandra, Virginia: American Diabetes Association; 1993.
4. Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. New Engl J Med 1993;329:977-86.

5. UK Prospective Diabetes Study (UKPDS) Group. Effect of intensive blood-glucose control with metformin on complications in overweight patients with type 2 diabetes (UKPDS 34). Lancet 1998;352:854-65.
6. UK Prospective Diabetes Study (UKPDS) Group. Intensive blood-glucose control with sulphonylureas or insulin compared with conventional treatment and risk of complications in patients with type 2 diabetes (UKPDS 33). Lancet 1998;352:837-53.
7. Strand L. Re-visioning the profession. J Am Pharm Assoc 1997;37:474-8.
8. Hepler C, Strand L. Opportunities and responsibilities in pharmaceutical care. Am J Hosp Pharm 1990;47:533-43.
9. Hatoum H, Akhras K. A 32 year literature review on the value and acceptance of ambulatory care provided by pharmacists. Ann Pharmacother 1993;27:1106-19.
10. Knowlton C, Thomas O, Williamson A, Gammaitoni A, Kirchain W, Buttaro M, et al. Establishing community pharmacy-based anticoagulation education and monitoring programs. J Am Pharm Assoc 1999;39:368-74.
11. Foss M, Schoch P, Sintek C. Efficient operation of a high-volume anticoagulation clinic. Am J Health-System Pharm 1999;56:443-9.
12. Jaber L, Halapy H, Fernet M, Tummalapalli S, Diwakaran H. Evaluation of a pharmaceutical care model on diabetes management. Ann Pharmacother 1996;30:238-43.
13. The DCCT Research Group. Reliability and validity of a diabetes quality-of-life measure for the Diabetes Control and Complications Trial (DCCT). Diabetes Care 1988;11:725-32.
14. Diabetes Control and Complications Trial Research Group. Influence of intensive diabetes treatment in quality of life outcomes in the Diabetes Control and Complications Trial. Diabetes Care 1996;19:195-203.
15. Khaw K, Wareham N, Luben R, Bingham S, Oakes S, Welch A, Day N. Glycated haemoglobin, diabetes, and mortality in men in Norfolk cohort of European Prospective Investigation of Cancer and Nutrition (EPIC-Norfolk). BMJ 2001;322:1-6
16. Miller LG. Herbal medicines. Arch Intern Med 1998;158:2200-11.
17. Myerscough M. Herbal remedies. How much do you know? Austr Fam Phys 1998;27:1037-40.
18. Gill G, Redmond F, Garratt F, Paisey R. Diabetes and alternative medicine: cause for concern. Diabetic Med 1994;11:210-13.
19. Shenfield G. What should we be doing about complementary therapies? Curr Ther 1999;August:7-11.
20. Angell M, Kassirer J. Alternative medicines — the risks of untested and unregulated remedies. New Engl J Med 1998;339:839-41.
21. Gori M, Campbell RK. Natural products and diabetes treatment. Diab Educator 1998;24:201-8.
22. Drew A, Myers S. Safety issues in herbal medicines: implications for the health professions. Med J Austr 1997;166:538-41.
23. Riley RH, Sutherland DM, Duffy TM. Herbal medicines. Austr J Hosp Pharm 1999;29:346.

Date article received 3.5.01; returned to author for revision 21.8.01; accepted for publication 19.2.02